CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 4: Linear Inequations (In One Variable)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Solve $2x + 3 < 11$, $x \in N$.
2. Solve $5 - 2x \geq 1$, $x \in W$.
3. Solve $-4x > 20$, $x \in Z$.
4. Solve $\frac{2x - 1}{3} \leq 3$, $x \in R$.
5. Solve $7 - 3x < 2x - 8$, $x \in R$, and say how it is shown on a number line.
6. Find the smallest integer x for which $3x - 4 > 10$.
7. Solve $-5 \leq 2x - 3 < 7$, $x \in Z$.
8. Solve $1 < 3x + 4 \leq 13$, $x \in R$.
9. If $A = \{x : 3x + 1 \geq 7,\ x \in R\}$ and $B = \{x : 4x - 5 < 15,\ x \in R\}$, find $A \cap B$.
10. Solve $x - 2 \leq 2x + 1 < 9$, $x \in I$.

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Answer Key

1. $x < 4$; $\{1, 2, 3\}$
2. $x \leq 2$; $\{0, 1, 2\}$
3. $x < -5$; $\{\ldots, -8, -7, -6\}$
4. $x \leq 5$
5. $x > 3$: hollow circle at 3 and a thick line to the right with an arrow
6. $x > 4\frac{2}{3}$, so 5
7. $-1 \leq x < 5$; $\{-1, 0, 1, 2, 3, 4\}$
8. $-1 < x \leq 3$
9. $\{x : 2 \leq x < 5,\ x \in R\}$
10. $-3 \leq x < 4$; $\{-3, -2, -1, 0, 1, 2, 3\}$

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